MERAKI - BIOS : RE-NEET 2026
TARGET 360/360 IN BIOLOGY
THREE [3] ALL INDIA FREE TESTS IN BIOLOGY
27.05.2026; 07.06.2026, 17.06.2026

PRACTICE TEST: REVISE AND RECALL: CELL: THE UNIT OF LIFE

1. Which of the following combinations of statements is CORRECT regarding cell structure and organelles?

(I) Ribosomes are membrane-bound organelles found in both prokaryotic and eukaryotic cells.

(II) Cytoplasm serves as the main arena of cellular activities in eukaryotic cells only.

(III) Prokaryotic cells lack membrane-bound organelles but still contain ribosomes.

(IV) Ribosomes in eukaryotic cells are found in the cytoplasm, mitochondria, and chloroplasts, as well as on rough ER.

(V) Centrosome is a non-membrane bound organelle found in animal cells that aids in cell division.

1. I, II and IV only
2. III, IV and V only
3. I, III and V only
4. II, III and IV only

[Page 88 - Recall -differences between eukaryotic / prokaryotic cells and animal / plant cells]

2. The correct order of layers in the bacterial cell envelope from outermost to innermost is:

1. Plasma membrane → Cell wall → Glycocalyx
2. Glycocalyx → Plasma membrane → Cell wall
3. Glycocalyx → Cell wall → Plasma membrane
4. Cell wall → Glycocalyx → Plasma membrane

[Page 90 - Recall - Mesosomes and chromatophores - very commonly asked in NEET]

3. What would MOST LIKELY happen if the hydrophobic tails of phospholipids were forced to face outward towards the aqueous environment instead of inward as they normally do?

1. The membrane would become more rigid and selectively permeable

2. The membrane structure would become unstable as the nonpolar tails would be exposed to the aqueous environment, disrupting the bilayer

3. The polar heads facing inward would strengthen the membrane by interacting with cytoplasmic water

4. Cholesterol would compensate and maintain membrane stability without any disruption

[Page 93 - Challenger - application of the concept in NCERT]

4. Which of the following options correctly identifies the organelles that are not considered part of the endomembrane system in a eukaryotic cell?

A.	Endoplasmic reticulum	B.	Golgi complex		
C.	Lysosomes	D.	Vacuoles		
E.	Mitochondria	F.	Chloroplasts	G.	Ribosomes

1. E, F and G only
2. A, B, C and D only
3. B, C, D and E only
4. A, D, E and G only

[Page 95; Recall - what is the advantage of compartmentalisation in a Eukaryotic cell]

5. Consider the given two statements:

A: The Golgi apparatus remains in close association with the endoplasmic reticulum.

B. Golgi apparatus is an important site of formation of glycoproteins and glycolipids.

1. Both A and B are correct and B explains A

2. Both A and B are correct but B does not explain A

3. Only A is correct

4. Only B is correct

[Page 96; Recall - The role of Golgi as ‘post office’ - the cis and trans face]

6. Which one of the following combinations of features is correctly shared by both mitochondria and chloroplasts?

1. Presence of single membrane, circular DNA and cristae

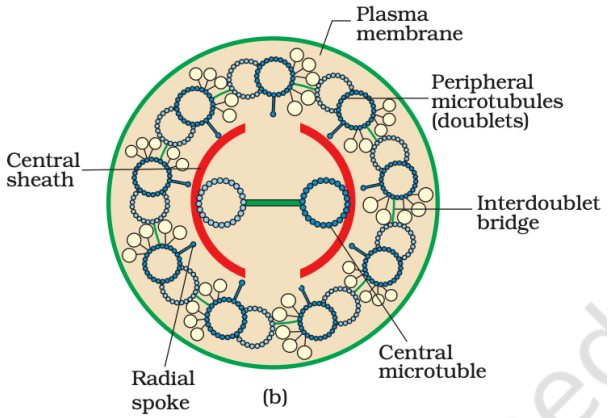
2. Presence of double membranes, genetic material and ribosome

3. Presence of thylakoids, matrix and hydrolytic enzymes

4. Presence of cisternae, oxysomes and chlorophyll pigments

[Pages 98/99; Recall - What explains their presence in Eukaryotic cells?]

7. The given figure shows the ultrastructure of:



1. Centriole
2. Centromere
3. Kinetochore
4. Cilia/flagella

[Page 99; Recall - what is a basal body?

8. Identify the correct statements:

I: The content of nucleolus is separated from the rest of the nucleoplasm as it is a membrane bound structure.

II: Nucleolus is a site for active ribosomal RNA synthesis.

III. Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.

1. Only I
2. Only II
3. Only I and II
4. Only II and III

[Page 100; Recall -What happens to the nucleolus at the time of mitosis]

9. The centromere is situated close to its end forming one extremely short and one very long arm in:

1. Metacentric chromosomes
2. Sub-metacentric chromosomes
3. Acrocentric chromosomes
4. Telocentric chromosomes

10. Ribosomes are the granular structures first observed under the electron microscope as dense particles by:

1. Robert Brown
2. Christian de Duve
3. George Palade
4. G. N. Ramachandran